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 Oefeningenreeks 2E nr. 26.12

$$\begin{aligned}
 \log_x \left(\frac{\log_2 6}{\log_{12} 6} + \log_2 \frac{1}{3} \right) &= \log_2 \left((2x)^{\frac{1}{6}} \right) \\
 \Leftrightarrow \log_x \left(\frac{(\log_2 6) \cdot (\log_2 12)}{\log_2 6} + \log_2 \frac{1}{3} \right) &= \log_2 \left((2x)^{\frac{1}{6}} \right) \\
 \Leftrightarrow \log_x \left(\log_2 12 + \log_2 \frac{1}{3} \right) &= \log_2 \left((2x)^{\frac{1}{6}} \right) \\
 \Leftrightarrow \log_x (\log_2 4) &= \log_2 \left((2x)^{\frac{1}{6}} \right) \\
 \Leftrightarrow \log_x 2 &= \frac{1}{6} \log_2 (2x) \\
 \Leftrightarrow 6 \log_x 2 &= \log_2 2x \\
 \Leftrightarrow 6 \frac{\log_2 2}{\log_2 x} &= 1 + \log_2 x \\
 \Leftrightarrow \frac{6}{\log_2 x} &= 1 + \log_2 x
 \end{aligned}$$

Stel $t = \log_2 x$

$$\begin{aligned}
 \Rightarrow \frac{6}{t} &= 1 + t \\
 \Leftrightarrow t^2 + t - 6 &= 0
 \end{aligned}$$

De oplossingen voor t zijn dan:

$$\begin{aligned}
 t_1 &= \frac{-1 + \sqrt{1 + 24}}{2} = -3 \\
 t_2 &= \frac{-1 - \sqrt{25}}{2} = 2
 \end{aligned}$$

De oplossingen voor x zijn dan:

$$\begin{aligned}
 \log_2 x_1 &= -3 \text{ of } \log_2 x_2 = 2 \\
 \Leftrightarrow x_1 &= \frac{1}{8} \text{ of } x_2 = 4
 \end{aligned}$$

References